1. What is a SOC?

SOC means Security Operations Center. It is a team of security people who watch a company's computers, servers and network all day and night. Their job is to find cyber attacks early and stop them before big damage happens. You can think of it like a control room with CCTV cameras, but for the company's network instead of a building.

A SOC watches logs and network traffic all the time and looks for strange or suspicious activity. When an alert comes, the team checks if it is a real attack or a false alarm. If it is real, they respond by stopping the attack, cleaning the system and helping it recover. After every incident they write a report and improve the security so the same problem does not happen again.

A SOC is needed because attacks can come at any time, even at night. If a company finds an attack late, the loss is bigger. A SOC helps to find and fix problems fast. A SOC has three main parts. The first is people, which means the security analysts. The second is process, which means the steps to follow when an alert comes. The third is technology, which means tools like SIEM, IDS, IPS and firewalls.

2. SOC Analyst Roles

A SOC team is divided into levels. Easy work goes to the first level, and difficult work goes to the higher levels. This way the most experienced people can focus on the serious problems.

The L1 analyst is the triage analyst and this is the entry level job. An L1 analyst watches the alerts and dashboards and checks if each alert is real or false. They follow ready-made steps called playbooks. If an alert is serious or they are not sure about it, they send it to L2.

The L2 analyst is the incident responder. They take the alerts that L1 sends and investigate them deeper to find out what happened. They find which systems are affected and how big the problem is. They also use threat information, like known bad IP addresses, to confirm the attack. After that they contain the attack, for example by blocking an IP address or isolating an infected computer.

The L3 analyst is the threat hunter or senior analyst, and is the most experienced person in the team. They handle the biggest and most difficult incidents. They also search for hidden threats that the tools did not catch, and they improve the detection rules and security tools. An easy way to remember the three levels is this: L1 finds it, L2 fixes it, and L3 hunts and improves.

3. Difference between SIEM, IDS and IPS

SIEM stands for Security Information and Event Management. It collects logs from many devices like servers, firewalls and applications and keeps them in one place. Then it looks at all the logs together and gives an alert if something looks suspicious. So SIEM shows the big picture of what is happening in the network. Examples of SIEM tools are Splunk, Wazuh and ELK Stack.

IDS stands for Intrusion Detection System. It watches the network traffic and gives an alert when it sees an attack pattern. It only detects and tells us about the attack, but it does not stop it. It sits on the side of the network and checks a copy of the traffic. Snort in detection mode is an example of an IDS, and this is what I used in this task.

IPS stands for Intrusion Prevention System. It is similar to IDS, but it sits in the middle of the traffic path. Because of this, it can detect the attack and also block it automatically. Examples of IPS are Suricata and Snort in inline mode.

The main difference is in what each one does. SIEM collects and compares logs from many sources, IDS only detects attacks, and IPS both detects and blocks attacks. These tools also work together. IDS and IPS send their alerts to the SIEM, and the SIEM joins these alerts with other logs, so the analyst can understand the full story of the attack.

4. Connection with my lab

In this week's lab I made a small SOC-style setup. I used an Ubuntu Server with Snort as an IDS. I wrote custom rules to detect ping (ICMP) and HTTP traffic. In a real SOC, these alerts would go to a SIEM and an L1 analyst would check them first.

5. Conclusion

A SOC is important for finding and stopping cyber attacks quickly. Analysts at L1, L2 and L3 work together, and tools like SIEM, IDS and IPS help them. SIEM collects and compares logs, IDS detects attacks, and IPS detects and blocks attacks. This week's lab gave me a basic idea of how detection works in practice.